



إمتحانات الدور الاول - ديسمبر 2025 م

المستوي الرابع - الدفعة (11) - قسم هندسة البرمجيات

التاريخ : 2025/12/20 م	اسم المادة: تطوير وتوثيق البرمجيات II
الزمن : 3 ساعات	استاذ المادة : د. كمال بشير

Answer all questions

Question 1. Definitions: Define the following terms.

1. Software Design Description (SDD)
2. Software Requirements Specification (SRS)
3. Architectural Design
4. Decomposition Description
5. Design Rationale
6. Data Dictionary
7. Component Design
8. Test Plan
9. Test Case
10. Traceability Matrix
11. Risk Analysis (in testing)
12. JavaDoc
13. @param tag
14. @throws tag
15. Checked Exception
16. Unchecked Exception
17. User Manual
18. Document Automation
19. Agile Documentation
20. Test Suite

Question 2. True/False Indicate T (True) or F (False):

Indicate whether the following statements are True (T) or False (F).

1. The SDD is the primary reference for code development and must contain all information a programmer needs. ()
2. The SRS describes how the system will be structured, while the SDD describes what the system should do. ()
3. A Test Plan describes the scope, approach, and schedule of intended testing activities. ()
4. The -private option in JavaDoc generates documentation for all classes and members, including private ones. ()
5. A Test Case typically includes elements like Test Case ID, Input Data, and Expected Results. ()

6. The purpose of a Traceability Matrix is to track features from the SRS to the test cases. ()
7. "Travel light" in Agile Documentation means you should create extensive documentation for every part of the system. ()
8. A Checked Exception in Java is a subclass of RuntimeException. ()
9. The main goal of Document Automation is to increase manual data entry and drafting time. ()
10. Risk Analysis in testing helps prioritize which features to test based on their likelihood and impact of failure. ()

Question 3: Direct Answers

Provide a short and direct answer to the following questions.

1. What are the two main stages of creating an SDD?
2. What are the three key elements that a true Document Automation System combines?
3. What is the primary purpose of the @return tag in JavaDoc?
4. List three of the five required elements of a Test Case.
5. What are two valid reasons for creating documentation according to Agile principles?
6. In the context of a User Manual, what is the main focus of a "tutorial approach"?
7. What is the key difference between a Test Plan and a Test Case?
8. Which JavaDoc tag would you use to indicate that a method should no longer be used?
9. What is the formula often suggested to evaluate the value of creating a document (involving C, R, U, F, T)?
10. What is one of the main benefits of using a Traceability Matrix?

Question 4: Discussion/Comparison/Drawing

1. Compare: What is the fundamental difference in focus between an SRS and an SDD?
2. Discuss: Why is "traveling light" considered a best practice in Agile Documentation?
3. Draw: Sketch a simple Test Case table structure with its core column headers.
4. Explain: What is the purpose of the "Design Rationale" section in an SDD?
5. Compare: What is the difference between a Checked and an Unchecked Exception in Java? Give an example of each.
6. Discuss: How does Document Automation improve profitability, as mentioned in the document?
7. Draw: Create a simple block diagram showing the three core aspects that a Document Automation System combines.
8. Explain: What is the role of a Data Dictionary in the context of a software design?
9. Compare: Contrast the "tutorial approach" and the "thematic approach" in organizing a User Manual.
10. Explain: How does performing a Risk Analysis assist in the test planning process?

Question 5:

Suppose you have an examination system for Karary University project:
Write the Software Design Specification (SDS)?

Good Luck!